

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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LAB REPORT

on

OPERATING SYSTEMS

Submitted by

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in

COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by ADARSH KP (1BM24CS015), who is Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026.

The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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2.	Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.	
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Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating System.
C04	Conduct practical experiments to implement the functionalities of Operating system.

Program1a

Question: Write a C program to simulate the following non-pre-emptive CPU scheduling algorithm to find turnaround time and waiting time.

- FCFS
- SJF (pre-emptive & Non-preemptive)

Soln: =>FCFS

```
#include <stdio.h>
#include <stdlib.h>

struct process {
    int id, at, bt, ct, tat, wt, rt;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
    }
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (p[j].at > p[j + 1].at) {
                struct process temp = p[j];
                p[j] = p[j + 1];
                p[j + 1] = temp;
            }
        }
    }
    int timepassed = 0, sumtat = 0, sumwt = 0, sumrt = 0;
    for (int i = 0; i < n; i++) {
        if (timepassed < p[i].at) {
            timepassed = p[i].at;
        }
        p[i].ct = timepassed + p[i].bt;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;

        timepassed += p[i].bt;

        sumtat += p[i].tat;
        sumwt += p[i].wt;
    }
    printf("\nID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
```

```

for(int i= 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].id, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}
printf("\nAverage TAT: %.2f\n", (float)sumtat / n);
printf("Average WT: %.2f\n", (float)sumwt / n);

return 0;
}

```

```

Enter total number of processes: 4
Process [1]
Arrival time: 0
Burst time: 2
Process [2]
Arrival time: 1
Burst time: 2
Process [3]
Arrival time: 5
Burst time: 3
Process [4]
Arrival time: 6
Burst time: 4

Process Arrival Time Burst Time Completion Time Turnaround Time Waiting Time
1         0         2         2             2             0
2         1         2         4             3             1
3         5         3         8             3             0
4         6         4        12             6             2

Average Turnaround Time: 3.50
Average Waiting Time: 0.75

```

SOLN: =>SJF(Preemptive)

```
#include<stdio.h>

struct process {
    int id, at, bt, ct, tat, wt, rt, rem;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
        p[i].rem = p[i].bt;
        p[i].rt = -1;
    }

    int completed=0, time=0, sumtat= 0, sumwt = 0, sumrt = 0;
    while (completed != n) {
        int idx = -1, minrem = 1e9;
        for (int i = 0; i < n; i++) {
            if(p[i].at<=time&& p[i].rem>0) {
                if(p[i].rem<minrem||(p[i].rem==minrem&&p[i].at<p[idx].at)) {
                    minrem = p[i].rem;
                    idx = i;
                }
            }
        }
        if (idx == -1) {
            time++;
            continue;
        }
        if (p[idx].rt == -1) {
            p[idx].rt = time - p[idx].at;
            sumrt += p[idx].rt;
        }
        p[idx].rem--;
        time++;
        if (p[idx].rem == 0) {
            completed++;
            p[idx].ct = time;
            p[idx].tat = p[idx].ct - p[idx].at;
            p[idx].wt = p[idx].tat - p[idx].bt;
            sumtat += p[idx].tat;
            sumwt += p[idx].wt;
        }
    }
}
```

```

printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
for(int i= 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].id, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}

printf("\nAverage TAT: %.2f\n", (float)sumtat / n);
printf("Average WT: %.2f\n", (float)sumwt / n);
return 0;
}

```

```

Enter number of processes: 5
Enter Arrival Time and Burst Time for P1: 2 1
Enter Arrival Time and Burst Time for P2: 1 5
Enter Arrival Time and Burst Time for P3: 4 1
Enter Arrival Time and Burst Time for P4: 0 6
Enter Arrival Time and Burst Time for P5: 2 3

PID    AT    BT    CT    TAT    WT
P1      2     1     3     1     0
P2      1     5    11    10     5
P3      4     1     5     1     0
P4      0     6    16    16    10
P5      2     3     7     5     2

Average Turnaround Time: 6.60
Average Waiting Time: 3.40
Process returned 0 (0x0)   execution time : 23.212 s
Press any key to continue.

```

SOLN: => SJF Non preemptive

```
#include<stdio.h>
struct process{
    int id,at,bt,ct,tat,wt,rt;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++){
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
    }

    int completed=0,time=0,sumtat=0,sumwt=0;
    int iscompleted[n];
    for(int i=0;i<n;i++) iscompleted[i]=0;

    while(completed!=n){
        int index=-1;
        int minbt=1e9;
        for(int i=0;i<n;i++){
            if(p[i].at<=time && !iscompleted[i]){
                if(p[i].bt<minbt){
                    minbt=p[i].bt;
                    index=i;
                }
            }
        }
        if(index!=-1){
            p[index].ct=time+p[index].bt;
            p[index].tat=p[index].ct-p[index].at;
            p[index].wt=p[index].tat-p[index].bt;
            p[index].rt=p[index].wt;
            sumtat+=p[index].tat;
            sumwt+=p[index].wt;
            time=p[index].ct;
            iscompleted[index]=1;
            completed++;
        } else {
            time++;
        }
    }

    printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
    for(int i=0;i<n;i++){
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",p[i].id,p[i].at,p[i].bt,p[i].ct,p[i].tat,p[i].wt);
    }
}
```



```
printf("\nAverage TAT: %.2f", (float)sumtat/n);  
printf("\nAverage WT: %.2f", (float)sumwt/n);  
}
```

```
Enter number of processes: 4  
Enter Arrival Time and Burst Time for P1: 0  
6  
Enter Arrival Time and Burst Time for P2: 0  
8  
Enter Arrival Time and Burst Time for P3: 0  
7  
Enter Arrival Time and Burst Time for P4:  
0  
3  
  
PID    AT    BT    CT    TAT    WT  
P1     0     6     9     9     3  
P2     0     8    24    24    16  
P3     0     7    16    16     9  
P4     0     3     3     3     0  
  
Average Turnaround Time: 13.00  
Average Waiting Time: 7.00  
Process returned 0 (0x0)  execution time : 25.867 s  
Press any key to continue.  
|
```

PROGRAM 1b

Question : Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time.

→ Priority (pre-emptive & Non-pre-emptive)

→ Round Robin (Experiment with different quantum sizes for RR algorithm)

SOLN: Priority Pre-emptive

```
#include <stdio.h>

struct process {
    int pid, at, bt, p, rt, ct, wt, tat;
};

int main() {
    int n, time = 0, completed = 0;
    printf("Enter the number of processes: ");
    scanf("%d", &n);
    struct process p[n];

    for (int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("Enter the arrival time, burst time, priority: ");
        scanf("%d%d%d", &p[i].at, &p[i].bt, &p[i].p);
        p[i].rt = p[i].bt;
    }

    while (completed < n) {
        int idx = -1, highest = 9999;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= time && p[i].rt > 0) {
                if (p[i].p < highest) {
                    highest = p[i].p;
                    idx = i;
                }
            }
        }
        if (idx != -1) {
            p[idx].rt--;
            time++;
            if (p[idx].rt == 0) {
                completed++;
                p[idx].ct = time;
                p[idx].tat = p[idx].ct - p[idx].at;
                p[idx].wt = p[idx].tat - p[idx].bt;
            }
        } else {
            time++;
        }
    }

    float totalwt = 0, totaltat = 0;
    printf("\nP\tA\tB\tP\tC\tT\tW\tT\n");
```

```

for(int i=0; i < n; i++) {
    totalwt+= p[i].wt;
    totaltat+= p[i].tat;
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt, p[i].p,
        p[i].ct, p[i].tat, p[i].wt);
}

printf("\nAverage waiting time : %.2f", totalwt / n);
printf("\nAverage turnaround time : %.2f\n", totaltat / n);
return 0;
}

```

```

Enter number of processes: 7
Enter AT, BT and Priority for P1: 0 8 3
Enter AT, BT and Priority for P2: 1 2 4
Enter AT, BT and Priority for P3: 3 4 4
Enter AT, BT and Priority for P4: 4 1 5
Enter AT, BT and Priority for P5: 5 6 2
Enter AT, BT and Priority for P6: 6 5 6
Enter AT, BT and Priority for P7: 10 1 1

```

PID	AT	BT	Pri	CT	WT	TAT
1	0	8	3	15	7	15
2	1	2	4	17	14	16
3	3	4	4	21	14	18
4	4	1	5	22	17	18
5	5	6	2	12	1	7
6	6	5	6	27	16	21
7	10	1	1	11	0	1

```

Average CT: 17.86
Average WT: 9.86
Average TAT: 13.71

```

SOLN : Non premetive

```
#include <stdio.h>

struct process {
    int id, at, bt, pr, ct, tat, wt, d;
};

int main() {
    int n, t = 0, c = 0, h;
    float tw = 0, tt = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);
    struct process p[n];

    for (int i = 0; i < n; i++) {

        p[i].id = i + 1;
        printf("Process %d\n", p[i].id);
        printf("Arrival time: ");
        scanf("%d", &p[i].at);
        printf("Burst time: ");
        scanf("%d", &p[i].bt);
        printf("Priority: ");
        scanf("%d", &p[i].pr);
        p[i].d = 0;
    }

    while (c != n) {
        h = -1;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= t && p[i].d == 0) {
                if (h == -1 || p[i].pr < p[h].pr) {
                    h = i;
                }
            }
        }

        if (h == -1) {
            t++;
        } else {
            t += p[h].bt;
            p[h].ct = t;
            p[h].tat = p[h].ct - p[h].at;
            p[h].wt = p[h].tat - p[h].bt;
            tw += p[h].wt;
            tt += p[h].tat;
            p[h].d = 1;
            c++;
        }
    }

    printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
```

```

        printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            p[i].id, p[i].at, p[i].bt, p[i].pr,
            p[i].ct, p[i].tat, p[i].wt);
    }

    printf("\nAverage Turnaround Time: %.2f", tt / n);
    printf("\nAverage Waiting Time: %.2f\n", tw / n);
    return 0;
}

```

```

Enter number of processes: 4

Process 1
PID: 1
Arrival Time: 0
Burst Time: 5
Priority: 2

Process 2
PID: 2
Arrival Time: 1
Burst Time: 3
Priority: 1

Process 3
PID: 3
Arrival Time: 2
Burst Time: 8
Priority: 4

Process 4
PID: 4
Arrival Time: 3
Burst Time: 6
Priority: 3

PID    AT    BT    PR    WT    TAT
1       0     5     2     0     5
2       1     3     1     4     7
3       2     8     4    12    20
4       3     6     3     5    11

Average Waiting Time = 5.25
Average Turnaround Time = 10.75

Process returned 0 (0x0)   execution time : 29.064 s
Press any key to continue.

```

SOLN : Round Robin

```
#include <stdio.h>

struct Process {
    int pid, at, bt, ct, tat, wt, rt;
};

int main() {
    int n, tq;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter Time Quantum: ");
    scanf("%d", &tq);

    struct Process p[n];
    int rem_bt[n];
    int readyQ[100];
    int front = 0, rear = 0;

    printf("Enter arrival times:\n");
    for(int i=0; i<n; i++) {
        scanf("%d", &p[i].at);
        p[i].pid = i+1;
    }

    printf("Enter burst times:\n");
    for(int i=0; i<n; i++) {
        scanf("%d", &p[i].bt);
        rem_bt[i] = p[i].bt;
        p[i].rt = p[i].bt;
        p[i].ct = p[i].tat = p[i].wt = 0;
    }

    int time = 0, completed = 0;
    int visited[n];
    for(int i=0; i<n; i++) visited[i]=0;

    for(int i=0; i<n; i++) {
        if(p[i].at == 0) {
            readyQ[rear++] = i;
            visited[i] = 1;
        }
    }

    while(completed < n) {
        if(front == rear) {
            time++;
            for(int i=0; i<n; i++) {
                if(!visited[i] && p[i].at <= time) {
                    readyQ[rear++] = i;
                    visited[i] = 1;
                }
            }
        }
    }
}
```

```

    }
    continue;
}

int idx = readyQ[front++];
if(rem_bt[idx] > tq) {
    time += tq;
    rem_bt[idx] -= tq;
    p[idx].rt = rem_bt[idx];
} else {
    time += rem_bt[idx];
    rem_bt[idx] = 0;
    p[idx].ct = time;
    p[idx].tat = p[idx].ct - p[idx].at;
    p[idx].wt = p[idx].tat - p[idx].bt;
    p[idx].rt = 0;
    completed++;
}

for(int i=0; i<n; i++) {
    if(!visited[i] && p[i].at <= time) {
        readyQ[rear++] = i;
        visited[i] = 1;
    }
}

if(rem_bt[idx] > 0) {
    readyQ[rear++] = idx;
}
}

printf("\nRRS scheduling:\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for(int i=0; i<n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}

float avg_TAT = 0, avg_WT = 0;
for(int i=0; i<n; i++) {
    avg_TAT += p[i].tat;
    avg_WT += p[i].wt;
}

printf("\nAverageturnaroundtime: %.2f ms\n", avg_TAT/n);
printf("Averagewaitingtime: %.2f ms\n", avg_WT/n);
return 0;
}

```

```
Enter number of processes: 5
Enter Arrival Time and Burst Time for P1: 0
5
Enter Arrival Time and Burst Time for P2: 1
3
Enter Arrival Time and Burst Time for P3: 2
1
Enter Arrival Time and Burst Time for P4: 3
2
Enter Arrival Time and Burst Time for P5: 4
3
Enter Time Quantum: 2
```

PID	AT	BT	CT	TAT	WT
P1	0	5	13	13	8
P2	1	3	12	11	8
P3	2	1	5	3	2
P4	3	2	9	6	4
P5	4	3	14	10	7

```
Average Waiting Time: 5.80
Average Turnaround Time: 8.60
```


PROGRAM 2

Question : Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

SOLN :

```
#include <stdio.h>

#define MAX 10

typedef struct {
int pid, at, bt, rt, ct, tat, wt, queue;
} Process;

int main() {
Process p[MAX];
int n, i, time = 0, completed = 0;
int tq = 2;

printf("Enter number of processes: ");
scanf("%d", &n);

for (i = 0; i < n; i++) {
printf("\nP%d\n", i + 1);
p[i].pid = i + 1;

printf("Arrival Time: ");
scanf("%d", &p[i].at);

printf("Burst Time: ");
scanf("%d", &p[i].bt);

printf("Queue (1-System, 2-User): ");
scanf("%d", &p[i].queue);

p[i].rt = p[i].bt;
}

printf("\nGantt Chart:\n");

while (completed < n) {

int executed = 0;

for (i = 0; i < n; i++) {
if (p[i].queue == 1 && p[i].rt > 0 && p[i].at <= time) {

executed = 1;

if (p[i].rt > tq) {
printf("P%d ", p[i].pid);
time += tq;
```

```

p[i].rt -= tq;
} else {
printf("P%d ", p[i].pid);
time += p[i].rt;
p[i].rt = 0;
p[i].ct = time;
completed++;
}
}
}

if (!executed) {
for (i = 0; i < n; i++) {
if (p[i].queue == 2 && p[i].rt > 0 && p[i].at <= time) {

executed = 1;

printf("P%d ", p[i].pid);
p[i].rt--;
time++;

if (p[i].rt == 0) {
p[i].ct = time;
completed++;
}

break;
}
}

if (!executed) {
time++;
}
}

float total_tat = 0, total_wt = 0;

printf("\n\nPID\tAT\tBT\tCT\tTAT\tWT\n");

for (i = 0; i < n; i++) {
p[i].tat = p[i].ct - p[i].at;
p[i].wt = p[i].tat - p[i].bt;

total_tat += p[i].tat;
total_wt += p[i].wt;

printf("P%d\t%d\t%d\t%d\t%d\t%d\n",

        p[i].pid, p[i].at, p[i].bt,
        p[i].ct, p[i].tat, p[i].wt);
}

printf("\nAverage TAT = %.2f", total_tat / n);
printf("\nAverage WT = %.2f\n", total_wt / n);

```

```
return 0;
}
```

```
Enter the total number of processes: 4

Process 1
Enter Arrival Time: 0
Enter Burst Time: 4
Enter Process Type (1 for System, 2 for User): 1

Process 2
Enter Arrival Time: 0
Enter Burst Time: 3
Enter Process Type (1 for System, 2 for User): 1

Process 3
Enter Arrival Time: 0
Enter Burst Time: 8
Enter Process Type (1 for System, 2 for User): 2

Process 4
Enter Arrival Time: 10
Enter Burst Time: 5
Enter Process Type (1 for System, 2 for User): 1

--- Scheduling Results ---
PID   Type  AT   BT   CT   TAT  WT
1     SYS   0    4    4    4    0
2     SYS   0    3    7    7    4
3     USR   0    8   15   15    7
4     SYS  10    5   20   10    5

Average Turnaround Time: 9.00
Average Waiting Time: 4.00

Process returned 0 (0x0)  execution time : 17.470 s
Press any key to continue.
|
```

PROGRAM 3

QUESTION : Write a C program to simulate Real-Time CPU Scheduling algorithms:

Rate- Monotonic

Earliest-deadline First

Proportional scheduling

SOLN: RateMonotonicScheduling

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
struct Process {
    int pid;
    int burst;
    int period;
    int remaining;
};

int gcd(int a, int b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct Process p[n];
    printf("Enter the CPU burst times:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &p[i].burst);
        p[i].pid = i + 1;
    }

    printf("Enter the time periods:\n");
    int hyperPeriod = 1;
    for (int i = 0; i < n; i++) {
        scanf("%d", &p[i].period);
        hyperPeriod = lcm(hyperPeriod, p[i].period);
    }

    printf("\nLCM = %d\n", hyperPeriod);
```

```

printf("\nRateMonotoneScheduling:\n");
printf("PID\tBurst\tPeriod\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\n", p[i].pid, p[i].burst, p[i].period);
}

double U = 0.0;
for (int i = 0; i < n; i++) {
    U += (double)p[i].burst/p[i].period;
}
double bound = n * (pow(2.0, 1.0/n) - 1);
printf("\n%.6f <= %.6f >= %s\n", U, bound, (U <= bound ? "true" : "false"));
printf("Scheduling occurs for %d ms\n", hyperPeriod);

for (int t = 0; t < hyperPeriod; t++) {
    int chosen = -1;

    for (int i = 0; i < n; i++) {
        if (t % p[i].period == 0) {
            p[i].remaining = p[i].burst;
        }
    }

    for (int i = 0; i < n; i++) {
        if (p[i].remaining > 0) {
            chosen = i;
            break;
        }
    }

    if (chosen != -1) {
        printf("%d ms onwards: Process %d running\n", t, p[chosen].pid);
        p[chosen].remaining--;
    } else {
        printf("%d ms onwards: CPU is idle\n", t);
    }
}

return 0;
}

```

```
Enter the number of processes: 2
Enter the CPU burst times:
20 35
Enter the time periods:
50 100
LCM=100

Rate Monotone Scheduling:
PID   Burst   Period
1      20      50
2      35      100

0.750000 <= 0.828427 =>true
Scheduling occurs for 100 ms

0ms onwards: Process 1 running
20ms onwards: Process 2 running
50ms onwards: Process 1 running
70ms onwards: Process 2 running
75ms onwards: CPU is idle

Process returned 0 (0x0)   execution time : 26.854 s
Press any key to continue.
|
```

SOLN : Earliest Deadline First

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
#define MAX_TASKS 10
```

```
typedef struct {
```

```
    int pid;
```

```
    int burst;
```

```
    int deadline;
```

```
    int period;
```

```
} Task;
```

```
int gcd(int a, int b) {
```

```
    while (b != 0) {
```

```
        int temp = b;
```

```
        b = a % b;
```

```
        a = temp;
```

```
    }
```

```
    return a;
```

```
}
```

```
int lcm(int a, int b) {
```

```
    return (a * b) / gcd(a, b);
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    Task tasks[MAX_TASKS];
```

```
    printf("Enter number of tasks: ");
```

```
    scanf("%d", &n);
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("\nTask %d:\n", i + 1);
```

```
        tasks[i].pid = i + 1;
```

```
        printf("Enter Burst Time: ");
```

```
        scanf("%d", &tasks[i].burst);
```

```
        printf("Enter Deadline: ");
```

```
        scanf("%d", &tasks[i].deadline);
```

```
        printf("Enter Period: ");
```

```
        scanf("%d", &tasks[i].period);
```

```
    }
```

```
    int SIMULATION_TIME = tasks[0].period;
```

```

for (int i = 1; i < n; i++) {
    SIMULATION_TIME=lcm(SIMULATION_TIME, tasks[i].period);
}

int remaining[MAX_TASKS] = {0};
int next_release[MAX_TASKS] = {0};
int abs_deadline[MAX_TASKS] = {0};
int active[MAX_TASKS] = {0};

printf("\nScheduling occurs for %d ms (LCM of periods)\n\n", SIMULATION_TIME);

for (int t = 0; t < SIMULATION_TIME; t++) {

    // Release jobs
    for (int i = 0; i < n; i++) {
        if (t == next_release[i]) {
            remaining[i] = tasks[i].burst;
            abs_deadline[i] = t + tasks[i].deadline;
            next_release[i] += tasks[i].period;
            active[i] = 1;
        }
    }

    int chosen = -1;
    int earliest = INT_MAX;

    for (int i = 0; i < n; i++) {
        if (active[i] && remaining[i] > 0) {
            if (abs_deadline[i] < earliest) {
                earliest = abs_deadline[i];
                chosen = i;
            }
        }
    }

    if (chosen != -1) {
        printf("%d ms: Task %d is running.\n", t, tasks[chosen].pid);
        remaining[chosen]--;

        if (remaining[chosen] == 0) {
            active[chosen] = 0;
        }
    } else {
        printf("%d ms : CPU is idle.\n", t);
    }
}

```



```
    return 0;
}
```

```
Enter the number of processes: 3
Enter capacities:
3 2 2
Enter deadlines:
7 4 8
Enter time periods:
20 5 10

LCM=20

Earliest Deadline First Scheduling:


| PID | Capacity | Deadline | Period |
|-----|----------|----------|--------|
| 1   | 3        | 7        | 20     |
| 2   | 2        | 4        | 5      |
| 3   | 2        | 8        | 10     |



Utilization = 0.750000

Scheduling:
0ms onwards: Process 2 running
2ms onwards: Process 1 running
5ms onwards: Process 3 running
7ms onwards: Process 2 running
9ms onwards: CPU is idle
10ms onwards: Process 2 running
12ms onwards: Process 3 running
14ms onwards: CPU is idle
15ms onwards: Process 2 running
17ms onwards: CPU is idle

Scheduling completed at 20 ms

Process returned 0 (0x0)   execution time : 11.491 s
Press any key to continue.
|
```

SOLN : Proportional Share Scheduling

```
#include<stdio.h>
#include <stdlib.h>
#include <time.h>
typedef struct
{
    char name[5]; int tickets; }
Process;
int main() {
    int n, total_tickets = 0; float total_T =0.0;
    printf("Enter the number of Processes: ");
    scanf("%d", &n);
    Process p[n];
    srand(time(NULL));
    for (int i = 0; i < n; i++)
    { printf("\nProcess %d:\n", i + 1); sprintf(p[i].name,
        "P%d", i + 1);
        printf("Tickets: ");
        scanf("%d", &p[i].tickets);
        total_tickets += p[i].tickets;
        total_T +=p[i].tickets;
    }
    printf("\n--- Proportional Share Scheduling ---\n");
    printf("Enter the Time Period for scheduling: ");
    int m;

    scanf("%d",&m);
    for (int i = 0; i < m; i++)
    {
        int winning_ticket = rand() % total_tickets + 1;
        int accumulated_tickets = 0;
        int winner_index;
        for (int j = 0; j < n; j++)
        {
            accumulated_tickets += p[j].tickets;
            if (winning_ticket <= accumulated_tickets)
            {
                winner_index = j; break;
            }
        }
        printf("Tickets picked: %d, Winner: %s\n", winning_ticket,
            p[winner_index].name);
    }
    for (int i = 0; i < n; i++)
    {
        printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n", p[i].name,
            ((p[i].tickets / total_T) * 100));
    }
```

```
}return 0; }
```

Output:

```
Enter number of processes: 3

Process 1
Enter weight: 10

Process 2
Enter weight: 20

Process 3
Enter weight: 30

Enter total CPU time: 5

--- Proportional Scheduling ---
Process P1 gets CPU time = 0.83 units
Process P2 gets CPU time = 1.67 units
Process P3 gets CPU time = 2.50 units
```

PROGRAM 4 :

QUESTION :

4a. Write a C program to simulate producer-consumer problem using semaphores

4b. Write a C program to simulate the concept of Dining Philosophers problem

SOLN : Bounded Buffer

```
#include <stdio.h>
#include <stdlib.h>
#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

int empty = BUFFER_SIZE;
int full = 0;

void produce() {
    if (empty == 0) {
        printf("\nBuffer is full! Cannot produce.\n");
        return;
    }

    int item;
    printf("Enter item to produce: ");
    scanf("%d", &item);

    buffer[in] = item;
    printf("Produced: %d\n", item, in);

    in = (in + 1) % BUFFER_SIZE;
    empty--;
    full++;
}

void consume() {
    if (full == 0) {
        printf("Buffer is empty! Cannot consume.\n");
        return;
    }

    int item = buffer[out];
    printf("Consumed: %d\n", item, out);

    out = (out + 1) % BUFFER_SIZE;
```

```

    full--;
    empty++;
}

int main() {
    int choice;
    printf("BufferSize: %d\n", BUFFER_SIZE);

    while (1) {
        printf("\n1.Produce\n2. Consume\n3. Exit\n");
        printf("Current State: [Full slots: %d | Empty slots: %d]\n", full, empty);
        printf("Enterchoice: ");

        if(scanf("%d", &choice) != 1) break;

        switch(choice) {
            case 1:
                produce();
                break;
            case 2:
                consume();
                break;
            case 3:
                printf("Exiting...\n");
                exit(0);
            default:
                printf("Invalid choice!\n");
        }
    }

    return 0;
}

```

```

Produced: 0 at buffer[0]
Consumed: 0 from buffer[0]
Produced: 1 at buffer[1]
Consumed: 1 from buffer[1]
Produced: 2 at buffer[2]
Produced: 3 at buffer[3]
Consumed: 2 from buffer[2]
Produced: 4 at buffer[4]
Produced: 5 at buffer[0]
Consumed: 3 from buffer[3]
Produced: 6 at buffer[1]
Produced: 7 at buffer[2]
Consumed: 4 from buffer[4]
Produced: 8 at buffer[3]
Produced: 9 at buffer[4]
Consumed: 5 from buffer[0]
Produced: 10 at buffer[0]
Consumed: 6 from buffer[1]
Produced: 11 at buffer[1]
Consumed: 7 from buffer[2]
Produced: 12 at buffer[2]
Consumed: 8 from buffer[3]
Produced: 13 at buffer[3]
Consumed: 9 from buffer[4]
Produced: 14 at buffer[4]
Consumed: 10 from buffer[0]
Produced: 15 at buffer[0]
Consumed: 11 from buffer[1]
Produced: 16 at buffer[1]
Consumed: 12 from buffer[2]
Produced: 17 at buffer[2]
Consumed: 13 from buffer[3]
Produced: 18 at buffer[3]

```

SOLN : Dining Philosopher

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
#define N 5 pthread_mutex_t
forks[N];      pthread_t
philosophers[N];

void* philosopher(void* num)
{
    int id = *(int*)num;
    int left = id; // left fork index
    int right = (id + 1) % N; // right fork index
    while (1)
    {
        printf("Philosopher %d is thinking.\n", id);
        sleep(1); // Thinking
        // To avoid deadlock, philosophers with even id pick left then right,
        // odd id pick right then left.
        if (id % 2 == 0)
        {
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id, left);
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id, right); }
        else
        {
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id, right); // Pick up left
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id, left);
        }
        printf("Philosopher %d is eating.\n", id);
        sleep(2); // Put down forks
        pthread_mutex_unlock(&forks[left]);
        pthread_mutex_unlock(&forks[right]);
        printf("Philosopher %d put down forks %d and %d.\n", id, left, right);
    }
    return NULL;
}

int main() {
    int i;
    int ids[N]; // Initialize forks (mutexes)
    for (i = 0; i < N; i++)
    { pthread_mutex_init(&forks[i], NULL);
```

```

ids[i] = i;
}
for (i = 0; i < N; i++){
pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
}

for (i = 0; i < N; i++){
pthread_join(philosophers[i], NULL); }
for (i = 0; i < N; i++){
pthread_mutex_destroy(&forks[i]);
}
return 0;
}

```

```

Buffer Size: 5
1. Produce
2. Consume
3. Exit
Current State: [Full slots: 0 | Empty slots: 5]
Enter choice: 1
Enter item to produce: 4
Produced: 4 at index 0

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 1 | Empty slots: 4]
Enter choice: 1
Enter item to produce: 8
Produced: 8 at index 1

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 2 | Empty slots: 3]
Enter choice: 2
Consumed: 4 from index 0

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 1 | Empty slots: 4]
Enter choice: 3
Exiting...

Process returned 0 (0x0)   execution time : 10.967 s
Press any key to continue.
|

```

PROGRAM 5

QUESTION : Write a C program to simulate Bankers algorithm and deadlock avoidance

SOLN : Banker algorithm

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int main() {
    int n, m;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &m);

    int allocation[MAX_PROCESSES][MAX_RESOURCES];
    int max[MAX_PROCESSES][MAX_RESOURCES];
    int need[MAX_PROCESSES][MAX_RESOURCES];
    int available[MAX_RESOURCES];

    printf("\nEnter Allocation Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Max Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    printf("\nEnter Available Resources:\n");
    for (int j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
```



```

        need[i][j]=max[i][j]-allocation[i][j];
    }
}

bool finish[MAX_PROCESSES]={false};
int safeSequence[MAX_PROCESSES];

int work[MAX_RESOURCES];

// Work = Available
for (int j = 0; j < m; j++) {
    work[j] = available[j];
}

int count = 0;

while (count < n) {
    bool found = false;

    for (int i = 0; i < n; i++) {

        if (finish[i] == false) {

            bool canExecute = true;

            for (int j = 0; j < m; j++) {
                if (need[i][j] > work[j]) {
                    canExecute = false;
                    break;
                }
            }

            if (canExecute) {

                for (int j = 0; j < m; j++) {
                    work[j] += allocation[i][j];
                }

                safeSequence[count] = i;
                count++;

                finish[i] = true;
                found = true;
            }
        }
    }
}

```

```

        if(!found) {
            break;
        }
    }

    if(count == n) {
        printf("\nSystem is in SAFE state.\n");
        printf("Safe Sequence: ");

        for(int i = 0; i < n; i++) {
            printf("P%d", safeSequence[i]);

            if(i != n - 1) {
                printf(" -> ");
            }
        }

        printf("\n");
    }
    else {
        printf("\nSystem is NOT in safe state.\n");
    }

    return 0;
}

```

```

Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Max Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in SAFE state.
Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2

```

SOLN : Deadlock avoidance algorithm

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int main() {
    int n, m;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &m);

    int allocation[MAX_PROCESSES][MAX_RESOURCES];
    int request[MAX_PROCESSES][MAX_RESOURCES];
    int available[MAX_RESOURCES];

    printf("\nEnter Allocation Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Request Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }

    printf("\nEnter Available Resources:\n");
    for (int j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }

    bool finish[MAX_PROCESSES];
    int work[MAX_RESOURCES];
    int safeSequence[MAX_PROCESSES];

    for (int j = 0; j < m; j++) {
        work[j] = available[j];
    }
```

```

}

for (int i = 0; i < n; i++) {
    bool zeroAllocation = true;

    for (int j = 0; j < m; j++) {
        if (allocation[i][j] != 0) {
            zeroAllocation = false;
            break;
        }
    }

    finish[i] = zeroAllocation;
}

int count = 0;

while (1) {
    bool found = false;

    for (int i = 0; i < n; i++) {

        if (finish[i] == false) {

            bool canExecute = true;

            for (int j = 0; j < m; j++) {
                if (request[i][j] > work[j]) {
                    canExecute = false;
                    break;
                }
            }

            if (canExecute) {

                for (int j = 0; j < m; j++) {
                    work[j] += allocation[i][j];
                }

                finish[i] = true;
                safeSequence[count] = i;
                count++;

                found = true;
            }
        }
    }
}

```

```

        if (!found) {
            break;
        }
    }

    bool deadlock = false;

    for (int i = 0; i < n; i++) {
        if (finish[i] == false) {
            deadlock = true;
            break;
        }
    }

    if (!deadlock) {
        printf("\nNoDeadlockdetected.\n");

        printf("Safe Sequence: ");
        for(inti=0;i<count;i++) {
            printf("P%d",safeSequence[i]);

            if (i != count - 1) {
                printf(" -> ");
            }
        }

        printf("\n");
    }
    else {
        printf("\nDeadlockdetected.\n");

        printf("Processesinvolvedin deadlock: ");

        for (int i = 0; i < n; i++) {
            if (finish[i] == false) {
                printf("P%d ", i);
            }
        }

        printf("\n");
    }

    return 0;
}

```

```
Enter number of processes: 5
Enter number of resources: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2
```

PROGRAM 6 :

QUESTION : Write a C program to simulate Memory Allocation (First Fit, Best Fit, Worst Fit)

SOLN : Memory Allocation

```
#include <stdio.h>

void firstFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                allocation[i] = j;
                blockSize[j] = 0;
                break;
            }
        }
    }

    printf("\nFirst Fit Allocation:\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

void bestFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int bestIdx = -1;

        for (int j = 0; j < m; j++) {
```

```

        if (blockSize[j] >= processSize[i]) {
            if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx]) {
                bestIdx = j;
            }
        }
    }

    if (bestIdx != -1) {
        allocation[i] = bestIdx;
        blockSize[bestIdx] = 0;
    }
}

printf("\nBest Fit Allocation:\n");
printf("Process No.\tProcess Size\tBlock No.\n");

for (int i = 0; i < n; i++) {
    printf("%d\t%d\t", i + 1, processSize[i]);

    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}

}

void worstFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int worstIdx = -1;

        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                if (worstIdx == -1 || blockSize[j] > blockSize[worstIdx]) {
                    worstIdx = j;
                }
            }
        }

        if (worstIdx != -1) {
            allocation[i] = worstIdx;
            blockSize[worstIdx] = 0;
        }
    }
}

```



```

    }

printf("\nWorst Fit Allocation:\n");
printf("Process No.\tProcess Size\tBlock No.\n");

for (int i = 0; i < n; i++) {
    printf("%d\t%d\t", i + 1, processSize[i]);

    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}
}

int main() {
    int m, n;

    printf("Enter number of memory blocks: ");
    scanf("%d", &m);

    int blocks[m], firstBlocks[m], bestBlocks[m], worstBlocks[m];

    printf("Enter sizes of memory blocks:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blocks[i]);
        firstBlocks[i] = blocks[i];
        bestBlocks[i] = blocks[i];
        worstBlocks[i] = blocks[i];
    }

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int processSize[n];

    printf("Enter sizes of processes:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
    }

    firstFit(firstBlocks, m, processSize, n);
    bestFit(bestBlocks, m, processSize, n);
    worstFit(worstBlocks, m, processSize, n);

    return 0;
}

```

```

Enter number of memory blocks
Enter size of each block:
100 500 200 300 600

Enter number of processes:4

Enter size of each process:
212 417 112 426

1. First Fit
2. Best Fit
3. Worst Fit
4. All Techniques
Enter your choice:4

```

```

--- FIRST FIT ---
Process No      Process Size      Block No
1               212             2
2               417             5
3               112             2
4               426            Not Allocated

--- BEST FIT ---
Process No      Process Size      Block No
1               212             4
2               417             2
3               112             3
4               426             5

--- WORST FIT ---
Process No      Process Size      Block No
1               212             5
2               417             2
3               112             5
4               426            Not Allocated

```

PROGRAM 7 :

QUESTION: Write a C program to simulate Page Replacement

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PAGES 100

void printFrames(int frames[], int num_frames) {
    printf("[");
    int count = 0;
    for (int i = 0; i < num_frames; i++) {
        if (frames[i] != -1) {
            printf("%d", frames[i]);
            count++;
        }
    }
    printf("]\n");
}

void fifoPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n---FIFOPageReplacement---\n");

    int frames[num_frames];
    for (int i = 0; i < num_frames; i++) frames[i] = -1;

    int page_faults = 0;
    int next = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {
                is_present = true;
                break;
            }
        }

        printf("Page %d -> ", page);

        if (!is_present) {
```

```

        frames[next] = page;
        next = (next + 1) % num_frames;
        page_faults++;
    }

    printFrames(frames, num_frames);
}
printf("TotalPageFaults(FIFO):%d\n",page_faults);
}

voidlruParamReplacement(intpages[],intnum_pages, int num_frames) {
    printf("\n---LRUPageReplacement---\n");

    int frames[num_frames];
    int last_used[num_frames];
    for (int i = 0; i < num_frames; i++) {
        frames[i] = -1;
        last_used[i] = -1;
    }

    int page_faults = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;
        int present_index = -1;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {
                is_present = true;
                present_index = j;
                break;
            }
        }

        printf("Page %d -> ", page);

        if (is_present) {

            last_used[present_index] = i;
        } else {

            page_faults++;

```

```

int empty_index = -1;
for(int j=0; j<num_frames; j++) {
    if (frames[j] == -1) {
        empty_index = j;
        break;
    }
}

if (empty_index != -1) {

    frames[empty_index] = page;
    last_used[empty_index] = i;
} else {

    int lru_index = 0;
    int min_time = last_used[0];
    for(int j=1; j<num_frames; j++) {
        if(last_used[j]<min_time){
            min_time = last_used[j];
            lru_index = j;
        }
    }
    frames[lru_index] = page;
    last_used[lru_index] = i;
}

}

printFrames(frames, num_frames);
}
printf("TotalPageFaults(LRU):%d\n", page_faults);
}

void optimalPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n---OptimalPageReplacement ---\n");

    int frames[num_frames];
    for(int i=0; i<num_frames; i++) frames[i] = -1;

    int page_faults = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;

        for(int j=0; j<num_frames; j++){

```

```

    if (frames[j] == page) {
        is_present = true;
        break;
    }
}

printf("Page %d -> ", page);

if (!is_present) {
    page_faults++;

    int empty_index = -1;
    for(int j=0; j<num_frames; j++) {
        if (frames[j] == -1) {
            empty_index = j;
            break;
        }
    }

    if (empty_index != -1) {
        frames[empty_index]=page;
    } else {

        int replace_index = -1;
        int farthest_use = -1;

        for(int j=0; j<num_frames; j++) {
            int next_use = -1;

            for(int k=i+1; k<num_pages; k++) {
                if(pages[k]==frames[j]) {
                    next_use = k;
                    break;
                }
            }
        }

        if (next_use == -1) {
            replace_index = j;
            break;
        }

        if(next_use>farthest_use) {
            farthest_use=next_use;
            replace_index = j;
        }
    }
}

```

```

        }

        frames[replace_index] = page;
    }
}

printFrames(frames, num_frames);
}
printf("TotalPageFaults (Optimal): %d\n", page_faults);
}

int main() {
    int num_pages, num_frames;
    int pages[MAX_PAGES];

    printf("Enter number of pages: ");
    if (scanf("%d", &num_pages) != 1 || num_pages <= 0 || num_pages > MAX_PAGES) {
        printf("Invalid number of pages.\n");
        return 1;
    }

    printf("Enter page reference string:\n");
    for (int i = 0; i < num_pages; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    if (scanf("%d", &num_frames) != 1 || num_frames <= 0) {
        printf("Invalid number of frames.\n");
        return 1;
    }

    fifoPageReplacement(pages, num_pages, num_frames);
    lruPageReplacement(pages, num_pages, num_frames);
    optimalPageReplacement(pages, num_pages, num_frames);

    return 0;
}

```

Enter number of pages:12

Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5

Enter number of frames:3

--- LRU Page Replacement ---

1	-1	-1
1	2	-1
1	2	3
4	2	3
4	1	3
4	1	2
5	1	2
5	1	2
5	1	2
3	1	2
3	4	2
3	4	5

Total Page Faults = 10

--- Optimal Page Replacement ---

1	-1	-1
1	2	-1
1	2	3
1	2	4
1	2	4
1	2	4
1	2	5
1	2	5
1	2	5
3	2	5
4	2	5
4	2	5

Total Page Faults = 7

--- FIFO Page Replacement ---

1	-1	-1
1	2	-1
1	2	3
4	2	3
4	1	3
4	1	2
5	1	2
5	1	2
5	1	2
5	3	2
5	3	4
5	3	4

Total Page Faults = 9